Firewall

LetsDefend SOC Analyst lesson | Asha Latha Panuganti | October 2026

Summary

Hands-on lab reading a FortiGate-format firewall log to identify the action taken, the source, the destination and the service involved.

Objectives

- Understand what a firewall log records and how allow/deny decisions are made.
- Extract action, source IP, destination IP and port, and service from a firewall log line.

Environment and tools

LetsDefend lab VM (Windows), Notepad++, FortiGate-format firewall log.

Method

- Opened the log and read each `name=value` field: `action`, `srcip`, `dstip`, `dstport`, `service`.
- Kept source and destination fields apart, and confirmed the port against the service and app fields.
- Checked bytes sent and policy type to confirm the outcome.

Findings

Log	Key fields observed	Finding
Firewall	srcip 192.168.68.12 on wan1 to dstip 192.168.68.34 port 143 (TCP); service/app IMAP; policytype local-in-policy; sentbyte 0	Inbound IMAP connection attempt was **denied** (action=deny) and no data was sent.

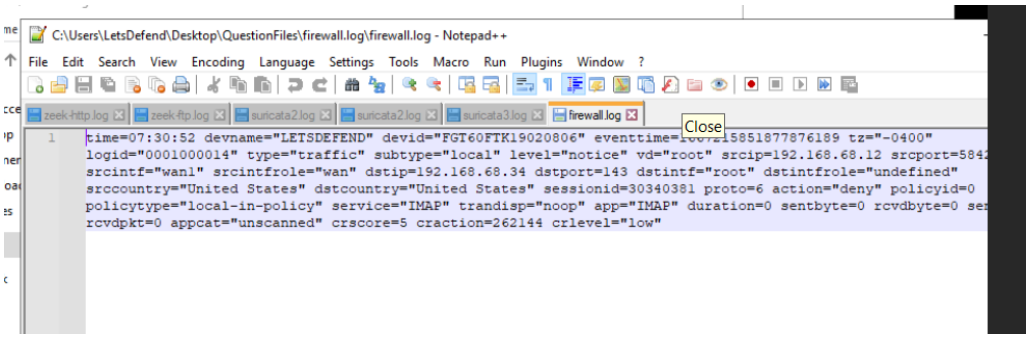


Figure 1: Firewall log showing action=deny for an IMAP connection (port 143).

Security observations

- **A deny is a good outcome, but still worth reading.** The IMAP attempt was blocked by a local-in policy with zero bytes sent. Repeated denies from one source can indicate scanning.
- **IMAP on port 143 is unencrypted.** Exposed mail services should use IMAPS (993).
- **Private source on a WAN interface.** The source is in a private range yet arrives on wan1. In production that would deserve a spoofing or misconfiguration check (likely a lab artifact here).

Skills demonstrated

Firewall log reading, TCP/IP and ports, allow/deny policy analysis, documenting findings.

Next steps

- Write a rule-review checklist for exposed services (IMAP, SSH, RDP).
- Forward firewall logs into Wazuh and alert on repeated denies from one source.